ECON 0100 | Exercise C2 | Taxes & Subsidies

Solutions

Answers and work are shown in the red blocks.

In recent years Hogsmeade has seen a proliferation of candy shops selling smoked toffees. The Demand (marginal benefit) curve and Supply (marginal cost) curve for smoked toffee can be represented by the following relationships.

$$D: P_B = 20 - \frac{1}{2}Q_D \qquad S: P_S = 2 + Q_S$$

Due to the metabolic issues among the younger generation of wizards in the community resulting from the consumption of these high glycemic index treats, the Ministry of Magic imposed a 3 Coin tax on the sale of smoked toffee.

Q1 | One Toffee Exchange

	a) A Single Exchange Buyer's MB: 16 Coins. Seller's MC: 12 Coins.	**b) A Single Exchange** Buyer's MB: 14 Coins. Seller's MC: 12 Coins.
With no tax, will they exchange?	_____	_____
With the tax, will they exchange?	_____	_____
Total Surplus to buyer and seller:	_____	_____
Tax paid to the Ministry of Magic:	_____	_____
Total Surplus (with the Ministry):	_____	_____
Deadweight Loss:	_____	_____

a) With no tax: Yes. With the tax: Yes. Total Surplus to buyer and seller: 1 Coin. Tax paid: 3 Coins. Total Surplus (with the Ministry): 4 Coins. Deadweight Loss: 0.

b) With no tax: Yes. With the tax: No. Total Surplus to buyer and seller: 0. Tax paid: 0. Total Surplus (with the Ministry): 0. Deadweight Loss: 2 Coins.

a) The buyer values the toffee at 16 and it costs the seller 12, so without the tax there is 4 of surplus to share and the trade happens. With the tax, 3 of that 4 goes to the Ministry, leaving 1 for the buyer and seller: still positive, so they still trade. Counting the Ministry's 3 as surplus (it is spent on something), total surplus is $1 + 3 = 4$, the same 4 as before. The tax moved surplus around but did not destroy any, so this exchange carries no deadweight loss.

b) Without the tax, $14 > 12$ and the trade happens, creating 2 of surplus. With the tax, the buyer's price would have to cover the seller's 12 plus 3 for the Ministry, 15, more than the 14 the buyer is willing to pay. So no trade: no surplus to the buyer and seller, and no tax paid either, since nothing is sold. The 2 this trade would have created is simply gone. That is the deadweight loss of a tax: it comes from the trades it prevents, not from the trades it taxes.

Q2 | Tax Equilibrium

Use a graph and algebra to discuss the impact of the tax on the market. Include how the welfare areas changed after the policy. No need to calculate welfare measures except DWL.

Pre-Tax Price ($\hat{P}$):	_____	Pre-Tax Quantity ($\hat{Q}$):	_____
Buyer's Price (P_b):	_____	Seller's Price (P_s):	_____
Post-Tax Quantity (Q^{Tax}):	_____	Dead Weight Loss (DWL):	_____
Tax Incidence on Buyers:	_____	Tax Incidence on Sellers:	_____

Who bears more of the tax? *buyers* · *sellers* · *equal*

Pre-Tax Price $\hat{P}$: 14 Coins. Pre-Tax Quantity $\hat{Q}$: 12 toffees. Buyer's Price P_b: 15. Seller's Price P_s: 12. Post-Tax Quantity Q^{Tax}: 10. Dead Weight Loss: 3 Coins. Tax Incidence on Buyers: 1. Tax Incidence on Sellers: 2. Who bears more: sellers.

Before the tax, buyers and sellers face one price:

$$20 - \tfrac{1}{2}Q = 2 + Q \quad \Rightarrow \quad Q = 12, \quad P = 14$$

With the tax, the buyer price sits 3 above the seller price. Slide the 3-Coin wedge in from the left until it gets stuck between the curves:

$$P_b = 3 + P_s \quad \Rightarrow \quad 20 - \tfrac{1}{2}Q = 3 + (2 + Q) \quad \Rightarrow \quad Q = 1$$

so $P_b = 20 - 5 = 15$ and $P_s = 2 + 10 = 12$.

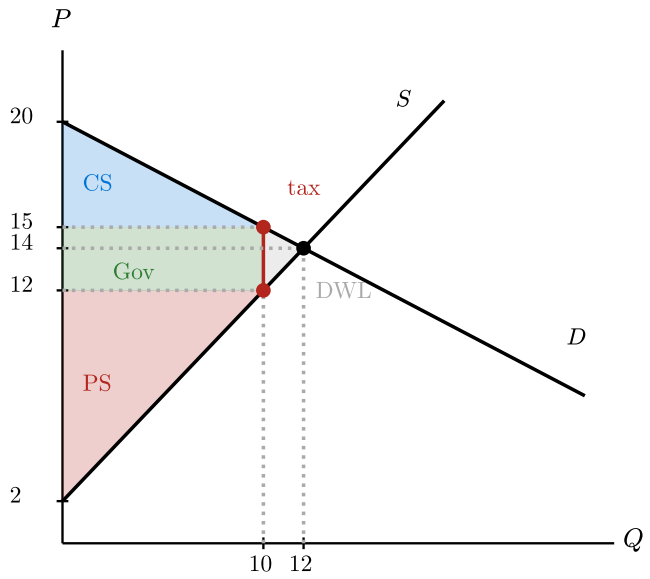


Figure 1: Tax equilibrium

Incidence: buyers pay $15 - 14 = 1$ more than before and sellers receive $14 - 12 = 2$ less, so sellers bear more of the tax. Supply is steeper (less elastic) than demand here, and the side that can adjust less bears more.

Welfare areas: consumer surplus shrinks to the triangle above 15, producer surplus to the triangle below 12, and the rectangle between the two prices over the 10 toffees still sold is the Ministry's

revenue, $3 \times 10 = 30$. The triangle between the curves over the 2 toffees no longer sold is the deadweight loss:

$$\text{DWL} = \tfrac{1}{2} \times 3 \times (12 - 10) = 3$$

Q3 | Subsidy Equilibrium

Now suppose that instead of the tax, the Ministry of Magic pays a 3 Coin subsidy on every smoked toffee sold. Use a graph and algebra to discuss the impact of the subsidy on the market.

Buyer's Price (P_b): _____ Seller's Price (P_s): _____

Post-Sub Quantity (Q^{Sub}): _____ Deadweight Loss (DWL): _____

Buyer's Price P_b: 13 Coins. Seller's Price P_s: 16 Coins. Post-Sub Quantity Q^{Sub}: 14 toffees. Deadweight Loss: 3 Coins.

A subsidy is the tax wedge turned around: the seller now receives 3 more than the buyer pays, so the wedge sits to the *right* of the no-policy equilibrium:

$$P_s = 3 + P_b \quad \Rightarrow \quad 2 + Q = 3 + (20 - \tfrac{1}{2}Q) \quad \Rightarrow \quad Q = 1$$

so $P_b = 20 - 7 = 13$ and $P_s = 2 + 14 = 16$.

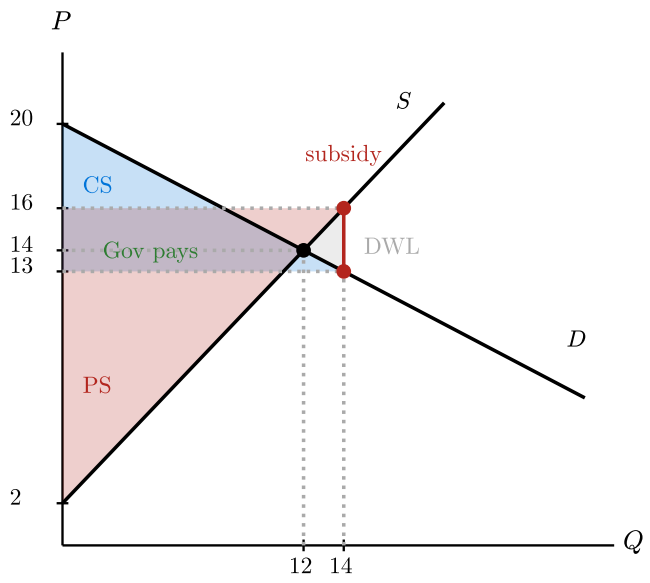


Figure 2: Subsidy equilibrium

Both sides are better off: buyers pay 1 less and sellers receive 2 more. But the Ministry pays $3 \times 14 = 42$, more than the two gains together, because the 13th and 14th toffees cost more to make (15 and 16) than buyers value them (13.5 and 13). They trade only because the Ministry pays the difference. The triangle between the curves over those 2 toffees is the deadweight loss:

$$\text{DWL} = \tfrac{1}{2} \times 3 \times (14 - 12) = 3$$